EDC BLOCK-003 (Derivation v6):
Collective Bulk Dimple and an Auto-Trapping Threshold

EDC Working Document

2026-02-02 — Program note / definitions [Open]; not a results paper

Abstract

This note introduces operational definitions for a "collective bulk dimple" formed by N nucleons on the EDC brane, and an associated auto-trapping threshold N_*. We define: the embedding function $\Xi_N(r)$ describing the brane deformation, its depth h_N and characteristic radius R_N; the effective test-particle energy $E_{\text{test}}(r;N)$ split into nuclear and environmental (geometric) contributions; and the energy gain $\Delta E(N) = E_{\text{test}}(\infty;N) - E_{\text{test}}(R_N;N)$ whose sign determines whether additional nucleons are energetically favored to join the cluster. The threshold N_* is defined by $\Delta E(N_*) = 0$. **What is not done:** No 5D back-reaction computation is performed; no normalization for κ_5^2 is assumed (the constant C issue remains); consequently, no quantitative prediction for N_* is made. **Conclusion:** BLOCK-003 remains [Open].

Contents

1 Notation Summary

Symbol	Definition	Tag
$\mathcal{M}^5$	5D bulk spacetime	[BL]
Σ^4	4D brane (our universe) embedded in $\mathcal{M}^5$	[BL]
$X^A = (x^\mu, \xi)$	Bulk coordinates; ξ = extra dimension	[BL]
$\xi = \Xi_N(r)$	Embedding function for N-nucleon cluster	[Open]
h_N	Dimple depth: $\max_r \Xi_N(r)$	[Open]
R_N	Characteristic cluster radius	[Open]
$\tau_{\mu\nu}^{(N)}$	Effective stress tensor for N nucleons	[Open]
κ_5^2	5D gravitational coupling ($= C\sigma^{-3/4}$, C unfixed)	[Open]
$\Delta E(N)$	Energy gain for test nucleon to join cluster	[Open]
N_*	Auto-trapping threshold: $\Delta E(N_*) = 0$	[Open]

Table 1: Notation used in this document. All [Open] quantities require 5D back-reaction computation for quantitative determination.

2 Minimal 5D Setup (Recap)

We recall the standard brane-world framework without re-deriving it.

Bulk: The 5D spacetime $\mathcal{M}^5$ with metric g_{AB} and Einstein-Hilbert action (plus Gibbons-Hawking-York boundary term):

$$S_{\text{bulk}} = \frac{1}{2\kappa_5^2} \int_{\mathcal{M}^5} d^5X \sqrt{-g^{(5)}} \left(R^{(5)} - 2\Lambda_5 \right) + \frac{1}{\kappa_5^2} \int_{\partial\mathcal{M}} d^4x \sqrt{-h}\, K$$

Brane: The 4D hypersurface Σ^4 with induced metric $h_{\mu\nu} = g_{\mu\nu} - n_\mu n_\nu$ (where n^A is the unit normal) and brane action:

$$S_{\text{brane}} = -\sigma \int_{\Sigma^4} d^4x \sqrt{-h} + S_{\text{matter}}$$

where σ is the brane tension and S_{matter} includes localized matter fields (nucleons as junction excitations in EDC language).

Status: This setup is standard [BL]. The [Open] element is the normalization of κ_5^2 (derivations v3–v5 established that the constant C in $\kappa_5^2 = C\sigma^{-3/4}$ remains unfixed).

3 The Israel Junction Conditions as a Bridge

The Israel junction conditions relate the discontinuity in extrinsic curvature across the brane to the brane stress-energy:

$$\boxed{[K_{\mu\nu} - K h_{\mu\nu}] = -\kappa_5^2 \left(\tau_{\mu\nu} - \frac{1}{3} \tau h_{\mu\nu} \right)} \tag{1}$$

where $[X] = X^+ - X^-$ denotes the jump across the brane, $K_{\mu\nu}$ is the extrinsic curvature, and $\tau_{\mu\nu}$ is the brane stress tensor (including tension σ and matter contributions).

3.1 The Bridge: Nucleon Binding → Geometry

Consider a compact cluster of N nucleons localized within a region of characteristic radius R_N on the brane. The key logical chain is:

1. **More nucleons in a compact region** $\Rightarrow$ larger effective stress $\tau_{\mu\nu}^{(N)}$ localized near $r \lesssim R_N$.
2. **Larger stress** $\Rightarrow$ larger jump in extrinsic curvature $[K_{\mu\nu}]$ via Eq. (1).
3. **Larger extrinsic curvature** $\Rightarrow$ deeper collective embedding $\Xi_N(r)$ into the bulk direction ξ.

This chain provides the formal bridge between nucleon binding (a brane-localized process) and bulk geometry (the "dimple").

Important: We do not claim a computed scaling law $h_N \propto f(N)$ here. The above is a qualitative/definitional statement about the direction of the effect. Quantitative determination requires a full 5D back-reaction calculation (currently [Open]).

4 Definitions: Embedding, Depth, and Cluster Size

Definition 1 (Collective Embedding Ansatz). *For a cluster of N nucleons centered at the origin, we parametrize the brane embedding by*

$$\xi = \Xi_N(r) \tag{2}$$

where r is the radial coordinate on the brane (3D spatial distance from the cluster center) and $\Xi_N(r) \geq 0$ with $\Xi_N(r) \to 0$ as $r \to \infty$.

Definition 2 (Dimple Depth).

$$h_N := \max_r \Xi_N(r) \tag{3}$$

The depth measures how far the brane "dips" into the bulk at the cluster location.

Definition 3 (Cluster Radius). *R_N is the characteristic radius enclosing most of the stress $\tau_{\mu\nu}^{(N)}$, operationally defined (e.g., radius containing 90% of the integrated stress, or the half-maximum width of Ξ_N).*

Scaling expectation [P]: On dimensional grounds, one might expect $R_N \sim N^{1/3} r_0$ where r_0 is a single-nucleon scale (e.g., $r_0 \sim 1\,\mathrm{fm}$). This is plausible but not derived here.

5 Auto-Trapping Criterion and Threshold Definition

5.1 Effective Potential for a Test Nucleon

Consider a test nucleon at radial position r in the presence of an existing N-nucleon cluster. Its total energy can be decomposed as:

$$E_{\text{test}}(r; N) = E_{\text{nuc}}(r; N) + E_{\text{env}}(r; N) \tag{4}$$

where:

- $E_{\text{nuc}}(r; N)$: nuclear contribution (strong force binding, standard nuclear physics) — largely understood.
- $E_{\text{env}}(r; N)$: environmental/geometric contribution induced by the collective embedding Ξ_N — the novel EDC component.

The geometric term E_{env} arises because the test nucleon, as a brane-localized excitation, couples to the local embedding geometry. A deeper dimple may provide an energetically favorable "niche."

5.2 Energy Gain and Auto-Trapping

Definition 4 (Energy Gain)**.**

$$\Delta E(N) := E_{\text{test}}(\infty; N) - E_{\text{test}}(R_N; N) \tag{5}$$

This is the energy difference between a test nucleon at infinity (unbound, flat brane) and one at the cluster edge R_N.

Interpretation:

- $\Delta E(N) < 0$: joining the cluster costs energy (requires external work; accretion is *not* spontaneous).
- $\Delta E(N) > 0$: joining the cluster releases energy (accretion is energetically favorable; "auto-trapping").

Definition 5 (Auto-Trapping Threshold)**.** *The critical nucleon number N_* is defined by:*

$$\boxed{\Delta E(N_*) = 0} \tag{6}$$

with $\Delta E(N) > 0$ for $N > N_*$.

For $N > N_*$, the collective dimple is "deep enough" that additional nucleons are energetically favored to join without external driving.

6 Schematic Illustration

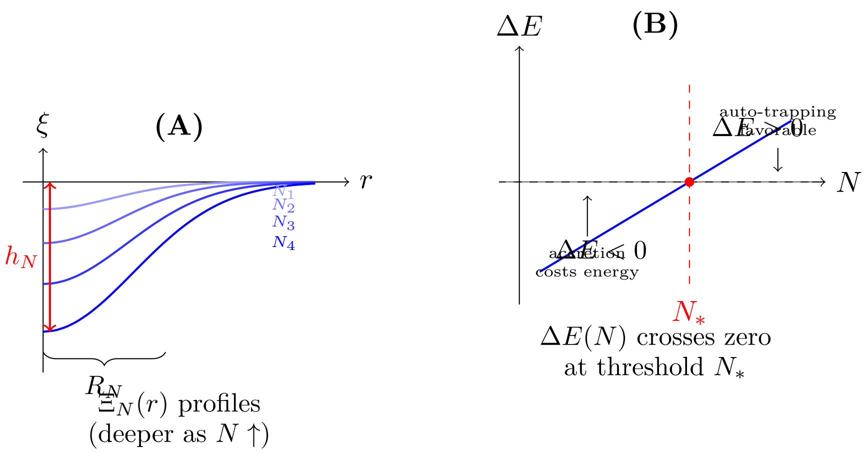

Figure 1: **(A)** Schematic embedding profiles $\Xi_N(r)$ for increasing nucleon number N; the dimple becomes deeper (h_N increases) and broader (R_N increases) with N. **(B)** The energy gain $\Delta E(N)$ transitions from negative (accretion unfavorable) to positive (auto-trapping) at the threshold N_* defined by $\Delta E(N_*) = 0$. *Schematic only; not to scale; definitions/hypothesis; no quantitative claim.*

Figure 1 illustrates the core concepts. Panel (A) shows how the embedding depth h_N grows with N; panel (B) shows the threshold behavior of $\Delta E(N)$.

7 What Remains Missing

Status: OPEN

The definitions and hypothesis above are **operational targets** for a future 5D calculation. The following elements are required to make quantitative predictions:

1. **5D back-reaction computation:** A calculation that maps $\tau_{\mu\nu}^{(N)} \to \Xi_N(r)$ through the full Einstein equations in the bulk. This would yield $h_N(N)$ and $E_{\text{env}}(r; N)$ explicitly.
2. **Normalization of κ_5^2:** The constant C in $\kappa_5^2 = C\sigma^{-3/4}$ must be fixed (see derivations v3–v5). Without this, even a complete back-reaction solution cannot predict N_* numerically.
3. **Specification of compactification scale L:** If G_N is to emerge correctly, the relation $G_N = \kappa_5^2/(6\pi L)$ requires L (see derivations v1–v2).

Conclusion: BLOCK-003 remains [Open]. This document provides definitions and a hypothesis; it does not constitute a derivation or a prediction.

8 Summary

Hypothesis 1 (Auto-Trapping Threshold — [Open])**.** *There exists a critical nucleon number N_* such that for $N > N_*$, the collective brane deformation $\Xi_N(r)$ induces an environmental energy contribution $E_{\text{env}}(r; N)$ making accretion of additional nucleons energetically favorable ($\Delta E > 0$) without external driving.*

What this note does:

- Defines Ξ_N, h_N, R_N, $\Delta E(N)$, N_* operationally.
- Articulates the Israel junction conditions as the bridge between nucleon binding and bulk geometry.
- Provides a schematic illustration (Figure 1).

What this note does not do:

- Compute $h_N(N)$ or $E_{\text{env}}(r; N)$ from 5D back-reaction.
- Fix the constant C in κ_5^2.
- Predict a numerical value for N_*.
- Close BLOCK-003.